Report Sheets
Lab 8: Rotational Motion

Name : ________________________ **Report Grade:_____________/70**

NetID : ________________________

Total Grade:_____________/100

Section : ________________________

Station #:_____ Today's teammates : ________________________________

❖ You are encouraged to have discussions with TA and other students, but you are required to do calculations and answer questions individually and independently.

Part I: Moment of inertia

Bare hanger's mass $=$ _______________ $\pm$ _______________ g

*For only one measurement, the error part is the measurement error.

Table R1:

	Trial 1	Trial 2	Trial 3	Mean
D of largest pulley (cm)				
D_{in} of ring (cm)				
Δ of ring (cm)				
Mass of ring (kg)	$M_{\text{ring}} =$ _______________ $\pm$ _______________ kg			

Table R2:

Case	Mass (g)	Angular acceleration α (rad/s^2)					
		Trial 1	Trial 2	Trial 3	Mean	σ	SEOM
1	50						
	100						
2	50						
	100						
3	50						
	100						

Check Box #1 [15]: The data sets are checked.

Part II: Energy conservation

Table R3:

Case	Final angular velocity ω_f (rad/s)					
	Trial 1	Trial 2	Trial 3	Mean	σ	SEOM
2						
3						

Part III: Angular momentum conservation

Table R4:

	Trial 1	Trial 2
ω_i (rad/s)		
ω_f (rad/s)		

Check Box #2 [15]: The data sets are checked.

Analysis I [20]: Moment of inertia

(1) From the measurements in **Table R1**, calculate and report the spindle pulley's radius r and the ring's inner radius R_{in} and outer radius R_{out}, all with the SI units (ignore the report of error). Finally, predict the ring's momentum of inertia $I_{\text{ring (predicted)}}$ (use the formula in Table 1 in **Manual** and express answers to 3 sig. figs.). [4]

- Spindle pulley's radius: $r = D/2 =$ __________________ m
- Ring's inner radius: $R_{\text{in}} = D_{\text{in}}/2 =$ __________________ m
- Ring's outer radius: $R_{\text{out}} = R_{\text{in}} + \Delta =$ __________________ m
- Prediction of Ring's moment of inertia:

$$I_{\text{ring (predicted)}} = \frac{1}{2} M_{\text{ring}} (R_{\text{in}}^2 + R_{\text{out}}^2) = \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ kg·m}^2$$

(2) Use the hanger's total mass m_h (= bare mass + added mass), spindle pulley's radius r [from **Analysis I**(1)], and measured angular acceleration α (from data in **Table R2**) to compute the torque τ exerted on the rotating object [hint: use Eq. (9) in **Manual**]. Finally obtain the moment of inertia as $I = \tau/\alpha$ and report I for each case as the average of the 50-g and 100-g rows. Complete **Table R5** below. [6]

Table R5: (Express answers to 3 sig. figs.)

Case	Mass (g)	m_h(kg)	α(rad/s^2)	τ (N·m)	I (kg·m^2)	Report
1	50					$I_1 =$ ______________ kg·m^2
	100					
2	50					$I_2 =$ ______________ kg·m^2
	100					
3	50					$I_3 =$ ______________ kg·m^2
	100					

(3) What should be the theoretical value of the ratio I_1/I_2 (hint: find information from Table 1 in **Manual**)? Discuss how the experimental ratio is compared with the theoretical one. [3]

Answer:

(4) What is the relation between I_2, I_3, and the ring's moment of inertia I_ring? Compute I_ring from the measured I_2 and I_3, and compare it with the predicted value in **Analysis I**(1). [3]

Answer:

(5) Please discuss (i) whether a rotating object's moment of inertia I should be dependent on or independent of the external torque τ and (ii) how your results have (roughly) confirmed or contradicted the hypothesis of (i). [4]

Answer:

Analysis II [10]: Energy conservation

(1) From the measured final angular velocity ω_f in **Table R3** and the disk's moment of inertia I in **Table R5**, compute the system's final kinetic energy $K_{\text{f,total}}$ [hint: It includes both translational and rotational kinetic energies. You may find Eq. (10) in **Manual** useful]. Compute the system's initial energy as the hanger's gravitational potential energy $U = m_{\text{h}}gh$ (m_{h} is the bare hanger's mass plus the added 0.1 kg mass), and then the energy loss rate. Complete **Table R6** below. [6]

Table R6: (Express answers to 3 sig. figs.)

Case	ω_f (rad/s)	I (kg·m^2)	$K_{\text{f,total}}$ (J)	U (J)	$\left\|\frac{K_{\text{f,total}} - U}{U}\right\| \times 100\%$
2					
3					

(2) Please discuss how your results have (roughly) confirmed or contradicted the energy conservation law. [4]

Answer:

Analysis III [10]: Angular momentum conservation

(1) From the angular velocities in **Table R4** and the moment of inertia in **Table R5**, analyze the system's momentum and energy loss upon the collision. Use the definition in **Manual** to compute the initial (final) angular momentum $L_i(L_f)$ and initial (final) rotational kinetic energy $K_i(K_f)$. (Note that the moment of inertial is I_2 before the collision and I_3 after the collision). Complete **Table R7** below. [6]

Table R7: (Express answers to 3 sig. figs.)

Trial #	Angular momentum (kg·m²/s)			Rotational kinetic Energy (J)		
	L_i	L_f	$\left\|\frac{L_f - L_i}{L_i}\right\| \times 100\%$	K_i	K_f	$\left\|\frac{K_f - K_i}{K_i}\right\| \times 100\%$
1						
2						

(2) Please discuss how your results have (roughly) confirmed or contradicted the conservation law regarding angular momentum and energy for a completely inelastic collision. [4]

Answer:

- **You can attach your calculation details (though it is not required). Attachments that demonstrate your understanding may gain you a partial credit.**
- **Please clean up, recover your work station, and sign out of the computer before leaving.**